

Sink-float separation - Proof of concept

Testing Document

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General description and purpose

Purpose of test

During an in-depth review of literature on plastic waste separation processes, the sink-float separation process emerged as the most promising in the setting of Blantyre. For most research toward plastic separation, the difficulty to translate the findings to real applications was criticized, since oftentimes virgin plastics are used which could vary drastically from post-consumer plastic waste in material properties and behavior (Wang et al., 2015). To validate the expected material behavior and to find the most suitable separation approach during the sink-float separation process, a proof-of-concept test series is conducted. These findings are then used to evaluate further separation steps.

Description of Test

For the test, the most common plastic waste types are used: PP, LDPE, HDPE, PS, PET, PVC. To allow for simple analysis of the results, all plastic types are color coded (plastic waste of different colors is used). For each item, the plastic type is validated by a handheld NIR spectrometer ([MicroPHAZIR PC](#)) and recorded together with its color. Additionally, the item's mass is recorded. Each item should weigh at least 60g. Alternatively, multiple identical items can be used for a single plastic type. Then, all items are separately crushed into plastic flakes using the [HoliShred V1](#). These flakes will be smaller than the ones stored in Blantyre, but according to research, larger flakes show better wettability (Pita & Castilho, 2017). Therefore, if it works in the lab with small flakes, it is expected to be transferable to the plastic waste in Blantyre. The resulting flakes are checked for purity and then weighted again.

For the tests, multiple batches of a mixture of the different plastic types are created. For each batch, 5 g of each plastic type are added to a container creating test batches of 30 g of mixed plastic waste. Twelve different proof-of-concept tests are run in small glasses or other transparent containers to which one of the plastic test batches is added, investigating differences in the separation processes after five minutes:

- a1. tap water
- a2. tap water, constant slow stirring (5 min)
- a3. tap water, vigorous stirring (3 min), let settle (2 min)
- b1. tap water, detergent (0.5 g per liter)¹
- b2. tap water, detergent (0.5 g per liter), constant slow stirring (5 min)
- b3. tap water, detergent (0.5 g per liter), vigorous stirring (3 min), let settle (2 min)
- c1. tap water, CaCl₂ (Calcium chloride, 1 g per liter)²
- c2. tap water, CaCl₂ (Calcium chloride, 1 g per liter), constant slow stirring (5 min)
- c3. tap water, CaCl₂ (Calcium chloride, 1 g per liter), vigorous stirring (3 min), let settle (2 min)
- d1. tap water, NaCl (Sodium chloride, 0.4 g per liter) as an alternative to CaCl₂
- d2. tap water, NaCl (Sodium chloride, 0.4 g per liter), constant slow stirring (5 min)
- d3. tap water, NaCl (Sodium chloride, 0.4 g per liter), vigorous stirring (3 min), let settle (2 min)

¹ Detergents / soap reduce the surface tension of water which should improve the wettability of hydrophobic plastic flakes.

² The addition of CaCl₂ has shown to increase the wettability of plastics (Pongstabodee et al., 2008).

Lastly, the results are evaluated: first, visually to see how well the separation of the colors has occurred, and then by measuring the two resulting fractions' total mass as well as the masses of the correctly and incorrectly assigned plastics. This allows to calculate the mass percentage / purity of the resulting fractions.

The expected / desired outcome is a clear separation between the low-density polymers (LDP) – PP, LDPE, HDPE – and the high-density polymers (HDP) – PS, PET, PVC.

Calculations

Separation medium concentration for salts CaCl_2 Dihydrate ($M = 147.01 \frac{\text{g}}{\text{mol}}, \frac{1\text{g}}{\text{l}}$) and NaCl ($M = 58.44 \frac{\text{g}}{\text{mol}}$):

$$C(\text{CaCl}_2 * 2\text{H}_2\text{O}) = \frac{n}{V} = \frac{m(\text{CaCl}_2 * 2\text{H}_2\text{O})}{M(\text{CaCl}_2 * 2\text{H}_2\text{O}) * V(\text{solution})} = \frac{1\text{g}}{147.01 \frac{\text{g}}{\text{mol}} * 1\text{l}} \approx 0.007 \frac{\text{mol}}{\text{l}}$$

$$C(\text{CaCl}_2 * 2\text{H}_2\text{O}) = C(\text{NaCl})$$

=> solution in 0.9 l water:

$$m(\text{CaCl}_2 * 2\text{H}_2\text{O}) = 0.9\text{g}$$

$$m(\text{NaCl}) = C(\text{NaCl}) * M(\text{NaCl}) * V(\text{solution}) = \frac{m(\text{CaCl}_2 * 2\text{H}_2\text{O})}{M(\text{CaCl}_2 * 2\text{H}_2\text{O})} * M(\text{NaCl}) \approx 0.4\text{g}$$

Plastic separation results:

$$\text{Purity } \varphi_{\text{float}}: \quad \varphi_{\text{float}} = m_{\text{float}_{\text{LDP}}} / m_{\text{float}}$$

$$\text{Recovery } R_{\text{float}}: \quad R_{\text{float}} = m_{\text{float}_{\text{LDP}}} / m_{\text{in}_{\text{LDP}}}$$

$$\text{Purity } \varphi_{\text{sink}}: \quad \varphi_{\text{sink}} = m_{\text{sink}_{\text{HDP}}} / m_{\text{sink}}$$

$$\text{Recovery } R_{\text{sink}}: \quad R_{\text{sink}} = m_{\text{sink}_{\text{HDP}}} / m_{\text{in}_{\text{HDP}}}$$

$$\text{Purity } \varphi_{\text{combined}}: \quad \varphi_{\text{combined}} = (\varphi_{\text{sink}} + \varphi_{\text{float}}) / 2$$

Schedule of test with all steps required

1. Acquire plastic waste of types PP, LDPE, HDPE, PS, PET, PVC in different colors.
2. For each item, scan the material with the NIR spectrometer and record its color and mass. (Each item should weigh at least 60g. Alternatively, multiple identical items can be used.)
3. Clean the HoliShred V1 of any remainders of old plastic. (DO NOT PUT HAND IN SHREDDER WHEN MOTOR IS PLUGGED IN!)
4. For each item, shred the item and collect the flakes in a separate container.
5. Record the resulting mass of flakes for each shredded item.
6. Prepare 3 batches of mixed test plastics which consists of 5 g of each plastic type (30 g total per batch).
7. Prepare 3 transparent containers as testing environments for test sequence **a1 – a3** with

- a1. **300 ml** of tap water
- a2. **300 ml** of tap water, magnetic stir bar, place on magnetic mixer
- a3. **300 ml** of tap water, magnetic stir bar, place on magnetic mixer
8. Set up camera to record the separation behavior over the test timeframe and start recording.
9. Add mixed test plastic batches to test containers and place the strainer on top. Start the mixer for batch **a2** at 300 rpm, and **a3** at 600 rpm, and start a timer of 5 minutes.
10. After 3 minutes, stop mixer of **a3**.
11. When the 5-minute timer is done, stop the mixer of **a2**.
12. For each batch separately, remove the floating fraction of the plastics of each container by skimming with a strainer, (for **b1 – d3**) rinse the flakes in the strainer with fresh water, and place them on a paper towel to dry.
13. For each batch separately, pour the rest of the container through the strainer over a sink to extract the sinking fraction of the plastics, (for **b1 – d3**) rinse the flakes in the strainer with fresh water, and place them on a paper towel to dry.
14. After the flakes have fully dried, record the total mass for each fraction in each batch.
15. Sort each fraction in each batch into LDP and HDP (using the color coding) and record the masses for each plastic type.
16. Calculate the recovery and purity for each of the 6 batches.
17. For each batch separately, mix the fractions to obtain the mix of plastic types of 30 g from step 6.
18. Repeat steps **7-16** once more to investigate changes in the process after wetting and drying the plastic flakes once.
19. Repeat steps **6-16** for the test sequences **b1 – b3**, **c1 – c3**, and **d1 – d3**. (This can be done in parallel, while the previous test sequence plastic is drying.) Prepare one batch of **1.5 l** of each solution before distributing it to the three containers to reduce deviation.
 - b1. **300 ml** of tap water, **0.15 g** of detergent
 - b2. **300 ml** of tap water, **0.15 g** of detergent, magnetic stir bar, place on magnetic mixer
 - b3. **300 ml** of tap water, **0.15 g** of detergent, magnetic stir bar, place on magnetic mixer
 - c1. **300 ml** of tap water, **0.3 g** of CaCl_2
 - c2. **300 ml** of tap water, **0.3 g** of CaCl_2 , magnetic stir bar, place on magnetic mixer
 - c3. **300 ml** of tap water, **0.3 g** of CaCl_2 , magnetic stir bar, place on magnetic mixer
 - d1. **300 ml** of tap water, **0.1 g** of NaCl
 - d2. **300 ml** of tap water, **0.1 g** of NaCl , magnetic stir bar, place on magnetic mixer
 - d3. **300 ml** of tap water, **0.1 g** of NaCl , magnetic stir bar, place on magnetic mixer

Risk analysis

Potential risks associated with this test

Shredding: Possible threat for hands (sharp blades) and eyes (plastic flakes flying off the shredder).

Intense stirring: Risk for plastic flakes or drops of the solution to get into the eyes or spilling.

CaCl_2 : The usage of solid CaCl_2 poses risks to eyes and skin due to its strong hygroscopic properties, possibly leading to irritation or burns, if in contact with the solid salt (“Calcium Chloride,” 2025).

Mitigation steps

Shredding: Safety glasses are worn during the whole shredding process. A rod is used if physical intervention with the shredder is necessary, never the hands.

Intense stirring: Safety glasses are worn during the experiments. Paper towels are prepared for the event of spills.

CaCl₂: Safety glasses and chemically resistant gloves are worn during the handling of the salt (s). The CaCl₂ container is only opened when preparing the solution and closed immediately afterwards.

Risks after taking mitigation steps

Shredding / Intense stirring: None.

CaCl₂: None, CaCl₂ is non-toxic when dissolved in water in small quantities (“Calcium Chloride,” 2025).

Tools and devices required

Tool	Check-in tick
1x safety glasses	X
1x pair of chemically resistant gloves	X
1x rod (wood or plastic)	X
Paper towels	X
1x laptop	X
1x camera (phone)	X
1x timer (laptop or phone can be used)	X
1x MicroPHAZIR PC	X
1x HoliShred V1	X
1x scale	X
6x container for different plastic types	X
3x container for mixed plastic test batches	X
3x transparent container for tests	X
1x large container (capacity ≥ 1 l)	X
Plastic waste, 60 g each (PP, LDPE, HDPE, PS, PET, PVC) in different colors	X (insufficient)
2x magnetic stir bar	X (one)

2x magnetic mixer	X (one)
3x strainer	X (two)
3.6 l tap water	X
0.45 g dish soap or other detergent	X
0.9 g CaCl ₂	X
0.4 g NaCl	X
1x tweezers	X
1x calculator	X

Testing Results

All recordings of the separation process and pictures of the results are available [here](#).

Test waste creation

Shredder clogged with PET. Therefore, scissors and wire cutters were used instead to cut flakes. → varying flake sizes, more similar to Blantyre waste

Waste item	Material (printed)	Material (scanned)	Color	Mass before shredding [g]	Mass after shredding [g]
Water bottle	PET	PET 0.983	Transparent, clear	27.6	54.1
Water bottle	PET	PET 0.978	Transparent, green	17.7	
Fruit tray	R-PET	PET 0.976	Transparent, clear	18.5	
Cookie tray	PET	Unidentified PET 0.302	Brown	6.5	
Cottage cheese cup	PS	PS 0.993	White	4.4	48.8
Yoghurt cup small x3	PS	PS 0.974	White	9.6	
Yoghurt cup large x4	PS	PS 0.951	White	34.9	
Cream cheese lid x2	PP	PP 0.998	Dark green, white print	8.3	13.0
Cream cheese lid	PP	PP 0.997	Dark green, white print	4.8	
Cracker packaging	PP	Unidentified PA 0.340	Brown-orange, inside silver	4.6	4.5 Not used
Detergent bottle	HDPE	PE 0.994	Blue	32.8	41.8
Bottle caps x6	HDPE	PE 0.997	Red	9.2	
Pipe	/	PVC 0.952	Gray	35.1	35
Vegetable bag (LDPE?)	/	PE 0.958	Transparent, red/brown	5.3	5.4

The resulting plastic flakes are depicted in Figure 1.

Since not for every plastic type enough material could be found, the mixed plastic test badge does not consist of 5 g each. Instead, a 1/20 of each plastic type is used for a mixed test plastic batch. Furthermore, the cracker packaging (PP) is not used in the test batch as its reflective side makes the visual differentiation difficult.



Figure 1: Prepared plastic flakes before test batch creation.

Separation test

Since only one magnetic stirrer was available, the vigorous stirring batches (a3, b3, c3, d3) were done by hand with a fork handle. The solution plastic mixture was stirred at a high velocity with periodic direction changes to create turbulences in the liquid.

Furthermore, the input mass measurements were done using a scale with an accuracy of 0.1 g. This leads to the difference in the theoretical mass of 9.6 g and the actual total input mass measured.

However, the resulting output masses for the sink and float fractions and their composition was done using a precise laboratory scale ([Mettler-Toledo NewClassic MS204/M](#)). This results in representative results for the purity, but not for the recovery values.

Sequence a

Run 1:

Test	Input masses [g]			Float-output masses [g]			Sink-output masses [g]		
a1	PP	LDPE	HDPE	PP	LDPE	HDPE	PP	LDPE	HDPE
	0.6	0.2	2.0	0.52	0.12	1.90	0	0	0
	PS	PET	PVC	PS	PET	PVC	PS	PET	PVC
	2.4	2.7	1.7	1.43	1.09	0.35	0.93	1.50	1.27
	Tot: 9.2 (theoretically 9.6)								
a2	PP	LDPE	HDPE	PP	LDPE	HDPE	PP	LDPE	HDPE
	0.6	0.2	2.0	2.72			0	0.01	0
	PS	PET	PVC	PS	PET	PVC	PS	PET	PVC
	2.4	2.7	1.7	0.12	0.03	0	6.58		
	Tot: 9.4 (theoretically 9.6)								
a3	PP	LDPE	HDPE	PP	LDPE	HDPE	PP	LDPE	HDPE
	0.6	0.2	2.0	2.59			0	0.003	0
	PS	PET	PVC	PS	PET	PVC	PS	PET	PVC
	2.4	2.7	1.7	0.19	0	0	6.39		
	Tot: 9.2 (theoretically 9.6)								

Run 2:

Test	Input masses [g]			Float-output masses [g]			Sink-output masses [g]		
a1	PP	LDPE	HDPE	PP	LDPE	HDPE	PP	LDPE	HDPE
	0.6	0.2	2.0	0.52	0.23	2.00	0	0.004	0
	PS	PET	PVC	PS	PET	PVC	PS	PET	PVC
	2.4	2.7	1.7	1.17	0.96	0.10	1.19	1.61	1.52
	Tot: 9.2 (theoretically 9.6)								
a2	PP	LDPE	HDPE	PP	LDPE	HDPE	PP	LDPE	HDPE
	0.6	0.2	2.0	2.63			0	0.01	0
	PS	PET	PVC	PS	PET	PVC	PS	PET	PVC
	2.4	2.7	1.7	0.21	0.07	0	6.45		
	Tot: 9.4 (theoretically 9.6)								
a3	PP	LDPE	HDPE	PP	LDPE	HDPE	PP	LDPE	HDPE
	0.6	0.2	2.0	2.63			0	0.001	0
	PS	PET	PVC	PS	PET	PVC	PS	PET	PVC
	2.4	2.7	1.7	0.27	0.01	0	6.28		
	Tot: 9.2 (theoretically 9.6)								

Any unexpected behaviors (clustering, foam formation, intermediate layers):

- a1 run 1: high purity, bad recovery (only what touches water)
- a1 run 1 (also other sequences for test 1): some material sinks as soon as the strainer touches the floating fraction when extracting

Sequence b

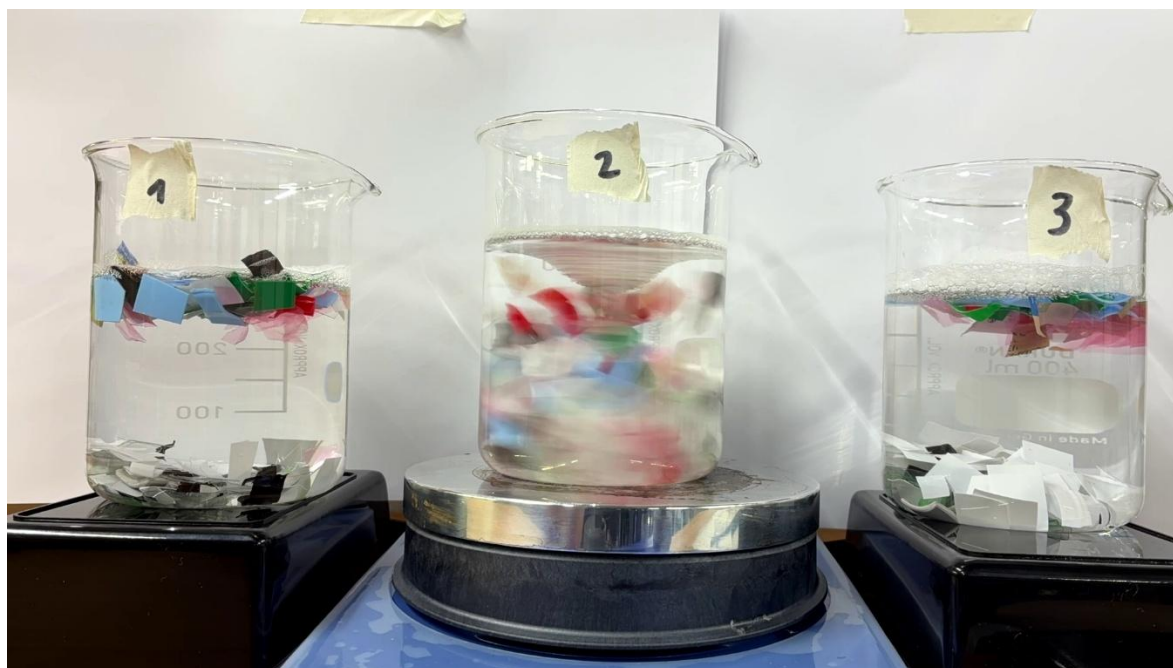


Figure 2: Separation setup for test sequence b.

Test	Input masses [g]			Float-output masses [g]			Sink-output masses [g]		
b1	PP	LDPE	HDPE	PP	LDPE	HDPE	PP	LDPE	HDPE
	0.6	0.2	2.0	3.36			0	0.0001	0
	PS	PET	PVC	PS	PET	PVC	PS	PET	PVC
	2.4	2.7	1.7	0.54	0.21	0	5.88		
	Tot: 10.0 (theoretically 9.6)								
b2	PP	LDPE	HDPE	PP	LDPE	HDPE	PP	LDPE	HDPE
	0.6	0.2	2.0	2.91			0	0.01	0
	PS	PET	PVC	PS	PET	PVC	PS	PET	PVC
	2.4	2.7	1.7	0.0001	0	0	6.43		
	Tot: 9.4 (theoretically 9.6)								
b3	PP	LDPE	HDPE	PP	LDPE	HDPE	PP	LDPE	HDPE
	0.6	0.2	2.0	2.61			0	0.003	0
	PS	PET	PVC	PS	PET	PVC	PS	PET	PVC
	2.4	2.7	1.7	0	0	0	6.70		
	Tot: 9.4 (theoretically 9.6)								

Any unexpected behaviors (clustering, foam formation, intermediate layers):

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Figure 3: Resulting fractions of sequence b.

Sequence c

Test	Input masses [g]			Float-output masses [g]			Sink-output masses [g]		
c1	PP	LDPE	HDPE	PP	LDPE	HDPE	PP	LDPE	HDPE
	0.6	0.2	2.0	2.64			0	0.0003	0
	PS	PET	PVC	PS	PET	PVC	PS	PET	PVC
	2.4	2.7	1.7	1.33	0.72	0.19	4.37		
	Tot: 9.3 (theoretically 9.6)								
c2	PP	LDPE	HDPE	PP	LDPE	HDPE	PP	LDPE	HDPE
	0.6	0.2	2.0	3.30			0	0	0
	PS	PET	PVC	PS	PET	PVC	PS	PET	PVC
	2.4	2.7	1.7	0.14	0.02	0	6.36		
	Tot: 9.9 (theoretically 9.6)								
c3	PP	LDPE	HDPE	PP	LDPE	HDPE	PP	LDPE	HDPE
	0.6	0.2	2.0	2.68			0	0.003	0
	PS	PET	PVC	PS	PET	PVC	PS	PET	PVC
	2.4	2.7	1.7	0.11	0.03	0	6.54		
	Tot: 9.3 (theoretically 9.6)								

Any unexpected behaviors (clustering, foam formation, intermediate layers):

- c2 & c3: some LDPE foil sticks to PS pieces in sink fraction

Sequence d

Test	Input masses [g]			Float-output masses [g]			Sink-output masses [g]		
d1	PP	LDPE	HDPE	PP	LDPE	HDPE	PP	LDPE	HDPE
	0.6	0.2	2.0	2.69			0	0	0
	PS	PET	PVC	PS	PET	PVC	PS	PET	PVC
	2.4	2.7	1.7	1.10	0.72	0	4.60		
	Tot: 9.2 (theoretically 9.6)								
d2	PP	LDPE	HDPE	PP	LDPE	HDPE	PP	LDPE	HDPE
	0.6	0.2	2.0	2.62			0	0	0
	PS	PET	PVC	PS	PET	PVC	PS	PET	PVC
	2.4	2.7	1.7	0.05	0.08	0	6.45		
	Tot: 9.4 (theoretically 9.6)								
d3	PP	LDPE	HDPE	PP	LDPE	HDPE	PP	LDPE	HDPE
	0.6	0.2	2.0	2.79			0	0.01	0
	PS	PET	PVC	PS	PET	PVC	PS	PET	PVC
	2.4	2.7	1.7	0.06	0	0	6.41		
	Tot: 9.3 (theoretically 9.6)								

Any unexpected behaviors (clustering, foam formation, intermediate layers):

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Evaluation

The calculation of the purity and recovery values are done in MS Excel and saved [here](#). The results regarding purity are summarized in Table 1.

Table 1: Purity results sorted by combined purity.

Test	Run	Purity float	Purity sink	Purity combined
b3	1	100.00%	99.96%	99.98%
b2	1	100.00%	99.84%	99.92%
d3	1	97.89%	99.84%	98.87%
c2	1	95.38%	100.00%	97.69%
d2	1	95.27%	100.00%	97.64%
c3	1	95.04%	99.95%	97.49%
a2	1	94.77%	99.85%	97.31%
a3	1	93.17%	99.95%	96.56%
a3	2	90.38%	99.98%	95.18%
a2	2	90.38%	99.85%	95.11%
b1	1	81.75%	100.00%	90.88%
d1	1	59.65%	100.00%	79.82%
a1	2	55.22%	99.91%	77.56%
c1	1	54.10%	99.99%	77.05%
a1	1	46.95%	100.00%	73.48%

All sink fraction results show a high purity of $> 99.84\%$. The float fractions on the other hand show a large difference in purity, highly dependent on the process. When no stirring is involved in the process, the float fraction purity lies between 46.95 and 81.75 %. When normal tap water is used without the addition of any surfactant with slow or vigorous stirring, the purity of the float fraction lies between 90.38 and 94.77 %. Out of the other processes – slow or vigorous stirring with the addition of surfactants – with a purity range of 95.04 to 100%, the processes involving the use of detergent shows the highest purities. When calculating a combined (averaged) purity of the float and sink fractions for each test, the slow and vigorous stirring of the detergent-water solution are established as the superior processes.

Due to the low accuracy of the scale used for the input plastic mixtures, the calculated recovery values are only of limited use. This is no concern since no rejection fraction is created during this separation process (leading to no unassigned plastic flakes) and, therefore, the purity measurement conveys this information simultaneously.

When comparing run 1 and run 2 of the sequence a, no clear pattern can be observed. This indicates no change of the plastic properties after a cycle of wetting and drying.

In conclusion, a **combination of stirring and water treated with detergent as a surfactant** are preferable. Next to improving the separation process, the detergent has the added benefit of acting as an additional cleaning step in the plastic waste processing.

The limited accuracy of the test procedure does not allow for a clear advantage of b2 or b3 over the other. This will be used as a degree of freedom in the prototype design.

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